

17.4 Maia: A Reticule-Scale AI Accelerator

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Abstract

In Paper 17.4, the architecture and implementation of Microsoft's MAIA AI silicon, a reticule-scale 750W AI SoC, is presented. Innovation across power delivery, thermal management,

physical design, validation, and packaging is detailed, resulting in a scalable approach to high-performance AI acceleration.

Introduction: Maia 200 is Microsoft's second-generation custom AI accelerator, engineered for large language model workloads. Compared to Maia 100, Maia 200 achieves over 3× higher HBM bandwidth and compute throughput, while scaling network bandwidth by more than 2× (see Fig. 17.4.1 for detailed specifications). Leveraging a vertically integrated approach across the system, networking, and software, Maia 200 is co-designed to deliver leading performance-per-dollar for inference and synthetic data generation workloads.

Similar to its predecessor, Maia 200 is a reticule-sized die, fabricated on TSMC's 3nm process and integrated using CoWoS-S packaging. With a provisioned 750W SOC TDP power, Maia 200 overcomes significant implementation challenges across power delivery, thermal management, physical design, package integration, and validation—topics that will be addressed in the following sections.

Architecture background: Maia accelerators adopt hierarchical architecture. The basic building block is a "tile." Each tile integrates two primary compute engines—a Tile Tensor Unit (TTU) and a Tile Vector Processor (TVP)—fed by a private, multi-banked Tile SRAM (TSRAM, or L1 SRAM) and a tile DMA subsystem for data movement. The TTU performs high throughput matrix multiply and convolution, which supports a wide range of data types including narrow data types such as FP8/FP6/FP4 that are OCP MX compliant. The TVP is a highly programmable 256-way SIMD unit, optimized for BF16 while supporting many other data formats. Each tile further contains a Tile Control Processor (TCP). The code running on the TCP is the final output of the software stack and delivers work to the tile subsystem. Hardware semaphores are used to synchronize DMA, TVP and TTU operations.

Multiple tiles compose together to form a "cluster." Each cluster contains a large multi-banked Cluster SRAM (CSRAM, or L2 SRAM), a dedicated cluster DMA subsystem for moving data between cluster SRAM and HBM memory, and a dedicated cluster core for control and synchronization. The entire SoC is constructed with multiple instances of clusters. To improve yield, redundancy schemes are implemented for both tiles and SRAM.

Maia accelerators feature a highly optimized data movement infrastructure, centered around its Direct Memory Access (DMA) subsystem and hierarchical Network-on-Chip (NoC), both critical for achieving scalable performance in AI and data-intensive workloads. The DMA engine is architected for multichannel and high-bandwidth data transfers, and it supports 1D, 2D and 3D data strides to optimize ML workloads. It enables asynchronous movement of data between on-chip SRAM, high-bandwidth memory (HBM), and external interfaces, supporting multi-stream operations to maximize memory throughput and minimize compute idle times. The NoC is designed as a hierarchical, scalable interconnect fabric that provides high-bandwidth, low-latency communication across compute clusters and memory subsystems. The NoC supports both unicast and multicast transfers, crucial for distributing large tensor blocks and synchronizing parallel processing elements. To further enhance HBM efficiency, multiple narrow data types are supported as storage formats in both HBM and SRAM. Hardware-based data casting converts these storage types to compute types at line rate, ensuring seamless integration with compute pipelines.

Maia accelerators employ a high-performance, Ethernet-based interconnect to deliver scalable, low-latency communication across multiple accelerator nodes. This interconnect leverages 112G PAM4 SerDes for high-speed data transfer. Unlike many other AI solutions, Maia implements a unified backend network architecture. The Ethernet links are organized into two categories: fixed links, which connect neighboring accelerators directly, and switch links, which interface with an Ethernet switch. This topology minimizes system cost and power consumption while improving link utilization for its target workloads.

Chip Implementation: Maia 200 employs TSMC's 3nm process with a reticule-limited die size (~26×33mm²). The design includes a large number of tiles with high-power density. To address the thermal and power delivery challenges inherent to reticule-scale silicon, the chip resources are partitioned into multiple independent clock and power domains. Dynamic Voltage and Frequency Scaling (DVFS) allows adaptation to workload, minimizing overall power usage. Clock gating is deployed aggressively, both at the register-transfer level for minor blocks and at the macro level, enabling rapid sleep and wake cycles responsive to AI workload demands.

One of the unique features for Maia is the large on-die SRAM. This allows data-intensive tasks to be serviced with minimal off-chip traffic, which helps reduce HBM demand and improve power efficiency. To improve Power-Performance-Area (PPA), customized SRAM macros are used for L1 and L2 SRAM. Given the large size of SRAM, the SRAM supports ECC correction and periodic scrubbing functions.

Maia 200 includes 28 X4 PAM4 channels for interprocessor communication. Each X4 channel is implemented using an X4 SerDes macro. To provide a better channel, each via

and BGA landing were optimized to meet the target impedance as close as possible. The CoWoS-S TSV model was optimized by characterizing the TSVs in a test interposer. Finally, end-to-end SI simulation was done and COM scores were met. For the host-side interface, Maia 200 uses Gen6 PCIe interface using similar SI practices as described above.

Maia 200 uses six stacks of HBM3e, three on each side of the SoC. Interposer routing was optimized for optimum read- and write-eye with lower power.

Maia 200 is a reticule-size die designed for data center deployments, where high availability is crucial. To achieve this, multiple RAS (Reliability, Availability, and Serviceability) features are integrated: HBM is a major source of potential errors; therefore, advanced on-die ECC and CRC mechanisms are implemented for HBM interfaces. On-demand memory scrubbing is supported for both HBM and on-die SRAM to proactively detect and correct faults. Additionally, the SoC enforces strict error containment policies to prevent error propagation across the system.

Power Design: The optimum power and cooling levels for the SoC are determined by assessing the target workload performance in the Azure data center. The data center control plane configures the power and cooling levels into the SoC at run time. Power consumption is monitored and regulated by the SoC. The compute blocks operate on a dedicated voltage and frequency domain to enable power scaling. This also allows Maia to respond to on-die and system events requiring power consumption reduction. There are on-die thermal and power sensors for runtime overheat detection, power management, as well as telemetry.

Power Delivery: Design a PDN that can supply such a high TDP power was an interesting problem to solve. There were many voltage domains and allocating enough resources to each domain while maintaining the supply noise and ripple specs was challenging to solve. The chip uses 112G Ethernet interface, PCIe6, and HBM3e interfaces. Care was taken to keep sensitive supplies for these interfaces isolated from any other supplies, so that the noise injection from other noisy supplies is low. Also, we made sure there are appropriate amounts of DTCs allocated for these sensitive supplies. And the supply noise was verified in PI analysis.

At the die level, there were several steps taken from the very beginning. For all clock buffers, a concise decision was made to have extra power/ground straps and dedicated decaps. SHDMIM caps were judiciously allocated closer to macros that have higher activity.

Maia 200 uses CoWoS-S packaging technology. For better power delivery Deep Trench Capacitor (DTC) was used for most rails. The availability of DTC helped significantly reduce supply noise.

As the demand current is very high, land side decaps were necessary for power integrity. The decap locations were chosen based on two major factors: first, the BGAs that are removed to make space for decaps, without causing other BGAs to exceed the current limit from reliability perspective; and second, the location of the decaps being close enough to be effective to regions where the change in demand current is high.

The power grid in interposer was done with the priority that each ubump connection to C4 has minimum lateral routing in interposer. To achieve this, we used a custom cell that contains a predefined number of ubumps and is connected to a C4 that is at the center of the ubumps at the other side of interposer. This structure provided a very uniform and well controlled C4-to-ubump resistance.

Finally, at the board level, the PMIC devices were placed close to the SoC package, so that the board level I²R loss could be minimized.

During the design phase, PDN models were generated for SoC, interposer, package, and board. We also included our internally generated PMIC models to simulate the full PDN network to ensure the specification for each rail was met.

Voltage Droop Management: Voltage excursions due to activity changes are a significant consideration due to the implications to power consumption. The SoC's compute can transition between low and high activity levels on short time scales of hundreds of nanoseconds. This can cause significant voltage excursions and power loss. The SoC implements a combination of proactive and reactive excursion management schemes. The compute sub-blocks are turned ON and OFF in a staggered manner to manage the ramp rates. The total amount of compute turning ON and OFF in a given time period is also controlled to limit the ramp rates. The ramp rate controls are tuned to minimize performance impact. Additionally, on-die voltage droop detectors sense and throttle the activity rates if proactive controls are not sufficient to manage the excursions within acceptable limits.

Physical Design: The physical design implementation is a cutting-edge engineering achievement that exemplifies how advanced methodologies can optimize performance, area, power, and manufacturability within the stringent constraints of a reticle-limited scale. As AI accelerators demand unparalleled levels of compute density, memory bandwidth, and energy efficiency, the Maia SoC addresses these challenges with an innovative design approach.

The chip's construction is built on a framework of quadrant-based compute clusters interconnected by a centralized fabric, with memory, PCIe, and scale-out I/Os strategically positioned along the periphery along with integration simplification through using abutted design implementation. However, achieving the desired performance metrics posed significant challenges, including handling the multi-hierarchical, multi-initiated module (MIM) designs, routing-intensive and memory-dominated architecture, multi-clock and multi-voltage domain interface crossings, high-speed operation and the integration complexity of diverse I/Os such as PCIe, HBM, and scale-out interfaces. Central to the success of the Maia accelerator is its robust Power Distribution Network (PDN), which utilizes 19 metal layers, hierarchical grids, and strategically placed bump pads to ensure reliable current delivery, mitigate IR drop and electromigration, and maintain signal integrity across the die. The chip also employs a custom clocking implementation based on an H-tree topology to minimize clock latency, variation, and skew, enabling high-speed and low-power operation. A customized routing methodology, leveraging all the metal layers to match RC delays across interface bundles and forward clocking implementation by taking advantage of unilateral data transfer, was deployed to optimize both area and latency.

Early-stage thermal simulations played a crucial role in optimizing the placement of high-switching regions, preventing thermal hotspots and improving overall thermal efficiency. Following place-and-route, the SoC undergoes comprehensive signoff flows to validate its functional correctness, performance, and manufacturability. These include multi-mode, multi-corner (MMMC) timing analysis to ensure timing closure across diverse conditions, IR drop and electromigration analysis for power grid robustness, DRC and LVS checks for process compliance and design integrity, and antenna effect mitigation through dummy vias and metal fills to enhance yield and reliability.

These measures ensure that the design is both performant and manufacturable at scale. The integration of these state-of-the-art techniques enables the Maia accelerator to achieve industry-leading performance metrics, including high clock speeds, high compute density, maximum memory fabric bandwidth, and energy efficiency, all while maintaining robust functionality and reliability. As a result, the SoC not only sets a new benchmark in AI accelerator SoC development but also demonstrates how large-scale chip designs can overcome the challenges of reticle-limited architectures, making it a flagship innovation in the field.

Package Technology: Maia 200 package is an enabling technology for scalable, high-performance AI silicon. The package architecture utilizes a silicon interposer, enabling dense interconnect routing and facilitating direct-to-die connections for high-speed, co-packaged memory subsystems.

An advanced silicon interposer forms the electrical and mechanical backbone of the package. The interposer supports ultra-fine pitch micro-bump connections between the main die and adjacent high-bandwidth memory (HBM3E) stacks. The interposer is fabricated with multiple metal layers, each optimized for high-frequency signal integrity and low RC loss, supporting aggregate bandwidths in the multi-terabit-per-second range.

Robust power delivery networks are implemented both within the interposer and the package substrate. Dedicated power and ground planes are assigned to support the high-current demands of the reticle-scale die, minimizing IR drop across the die and HBM. Power/ground bumps are distributed strategically to correspond to areas with the highest current draw, such as tiles and memory controllers. Signal escape routing leverages the interposer's dense wiring capabilities and the package's multi-layer to facilitate reliable, low-crosstalk communication with external system components via high-speed SerDes and PCIe links.

In-package thermal sensors provide real-time temperature monitoring, enabling dynamic control over cooling systems and supporting the chip's adaptive power management features.

Fine-pitch ball grid arrays (BGA) connect the package to the system board, enabling compact footprint and robust mechanical stability. The overall package structure is engineered to manage stress induced by thermal cycling and power transients, with underfill and advanced encapsulants used to protect micro-bump regions and interposer connections. Extensive simulation and reliability testing—including thermal shock, vibration, and electrostatic discharge (ESD)—are conducted to ensure service lifetime in demanding data center operations. The Silicon, Package, Interconnect, board, system mechanics and thermals are all designed and optimized with the view of end-to-end optimization across the stack.

Validation Strategy: Maia 200 employs a staged pre- and post-silicon validation strategy encompassing Field Programmable Gate Arrays (FPGA) prototyping, emulation, wafer-level bring-up, packaged silicon bring-up, silicon validation and qualification, and rack-scale system testing. A standalone FPGA platform enabled early bring-up of SoC firmware and drivers, exercising substantial portions of the chip logic. The FPGA was also linked to silicon bring-up and production boards via adapters to validate PCIe interfaces and FW/SW in a production-like environment—critical for firmware readiness and a seamless transition to silicon.

Emulation complemented these efforts by supporting long-duration functional and performance stress suites that map directly to silicon, covering approximately 95% of validation without relying on the production software stack. Firmware and host drivers were also integrated on emulation, alongside validation of basic kernels and models for both functionality and performance. High-speed I/O initialization sequences were fully verified across simulation, emulation, and FPGA platforms. These measures ensured smooth deployment of the production firmware to stack upon silicon arrival. Stress tests for the production FW/SW stack were likewise exercised on emulation prior to silicon, accelerating readiness.

Conclusion: Maia 200 delivers a reticle-scale AI accelerator optimized for performance, energy efficiency, and manufacturability. Leveraging a vertically integrated approach across system and software design, Maia 200 achieves 3-to-5× performance-per-dollar improvement over Maia 100 on Azure's in-house generative inference workloads. This advancement sets a new benchmark for high-performance AI acceleration in hyperscale data center environments.

	Maia 100	Maia 200
Peak Dense Tensor TOPS	6bit: 3000	FP4: 10145
	9bit: 1500	FP8: 5072
	BF16: 800	BF16: 1268
Peak HBM BW (TB/s)	1.8TB/s	Up to 7TB/s
Num of HBM Stack	4	6
HBM Capacity (GB)	64	216
Host PCIe BW (GB/s)	32	64
PCIe Configuration	Gen4 x16/Gen5 x8	Gen6 x8
Backend Network Bandwidth (GB/s) (bi-directional)	1200	2800
Backend Network Configuration	12x400	28x400

Figure 17.4.1: Maia 100 and Maia 200 performance summary.